Smart Product Pricing: A Multi-Modal Deep Learning Approach with Gradient Boosting for E-Commerce Price Prediction

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***Abstract*—Price prediction for products on e-commerce is a challenging task owing to scale, multiple data modalities, and wide price ranges across categories. Most existing techniques using only meta-data fall short of utilizing the rich information in the product images and text. In this paper, we introduce Smart Product Pricing (SPP), which employs a multi-modal deep learning approach for predicting the price of a product by integrating product images and text. We introduce *price-aware feature extraction*, i.e., training of CNN-Transformer architecture from scratch for extracting features in an end-to-end manner and then fine-tuning the backbone under price supervision and finally replacing dense head with LightGBM gradient boosting. This framework is designed through three models: (1) end-to-end multi-modal neural network model with dense head (SMAPE 46.1%), (2) two-stage neural network model with LightGBM gradient boosting (SMAPE 31.2%), and (3) final multi-modal architecture incorporating PCA for dimensionality reduction from 2048 to 300 (SMAPE 19.73%, MAE Rs.4.82, R^2=0.77). Experiments were conducted using Mavericks Amazon ML challenge dataset of 75,000 products belonging to clothing, electronics, home items, and personal care evaluated on a 90/10 train/validation split (7,500 validation samples). Statistical significance was ensured through paired t-test (t=29.05, p<0.0001). Key limitations include single-image input and absence of structured category features.**

***Index Terms*—multi-modal learning, price prediction, e-commerce, LightGBM, EfficientNet, DistilBERT, PCA, gradient boosting, transfer learning**

I. Introduction

There are multiple e-commerce portals like Amazon, Flipkart, and Alibaba which have millions of products listed under different consumer categories. One of the major problems faced here is price prediction, that is, finding a good price for a given product according to its features. Small and medium sized sellers suffer from this problem because of lack of competitive pricing and experience in the field.

Improper pricing has a direct impact; low pricing results in low margins, and high prices result in reduced velocity of sales. Apart from this, price prediction also helps in detecting fraudulent activities, for example, selling of very cheap high value electronics, dynamic repricing and also helps in quality management of the market place. Other problems associated include extreme heterogeneity in products (price range of 'watch' can be anything between Rs.100 to Rs.5,00,000), missing metadata information from the seller and dynamic pricing due to seasonality and fluctuations in demand.

The current approach uses only the structured form of metadata which includes categorization of products, brands, dimension details, purchase history etc. However, there are other forms of information like product images and textual descriptions in unstructured format which complement the structured information and cannot be captured by it.

Research Gap: Despite a number of attempts to investigate multi-modal learning in vision-language settings, no research was done to apply it to price regression in e-commerce domain. All existing multi-modal pricing algorithms use either generic pretrained features without any task adaptation or end-to-end neural networks that fail to perform well on the essentially tabular fused embeddings problem. None of the previous works used price-supervised embeddings extraction together with PCA-regularized gradient boosting to solve the problem.

Thus, there is a need for SPP approach. This paper makes the following contributions:

- We design and test three progressive architectures that trace a natural performance trendline from 46.1% to 19.73% SMAPE on 75,000 Amazon India products.
- We propose *price-aware embedding extraction*: joint fine-tuning of CNN+Transformer encoders in price-supervised

manner and subsequent freezing to get specialized features for gradient boosting.
- We show that PCA compression from 2048 to 300 dimensions lowers the ratio between training and validation errors by reducing it from 4.89× to 1.63× and getting R^2=0.77 on 7,500 held-out samples.
- We conduct modality ablation analysis and statistical testing (t=29.05, p<0.0001) to verify improvements are not accidental.

II. Related Work

A. Price Estimation in E-commerce

Previous efforts for estimating the price of products have leveraged tabular data through linear regression, random forest, XGBoost, and LightGBM models [1]. Even though these methods are successful on highly curated datasets that possess plenty of information, they do not have the capability to leverage information within the images of the products and text descriptions. A critical limitation is that structured metadata is incomplete or absent for many listings, making these methods brittle in real marketplace scenarios. In more recent times, text encoders based on transformers have been used in price estimation tasks [9]; however, the image modality has not been considered, ignoring a key quality signal that cannot be fabricated by sellers.

B. Multi-Modal Learning

Multi-modal learning has been able to reach state-of-the-art results on various vision-language tasks, such as visual question answering [2], image captioning, and cross-modal retrieval. The power of contrastive vision-language pre-training has been shown by CLIP [11], while BLIP-2 [16] further advanced bootstrapped vision-language pre-training with frozen encoders. Vision-and-Language pre-training methods like VL-BERT [12] and Multi-modal Transformers have been tried for product understanding tasks. However, these approaches are primarily designed for classification and retrieval tasks, not regression with skewed price distributions. Fusion techniques commonly used are early fusion (by concatenating features), late fusion (by averaging predictions) and cross attention. Our approach uses early fusion through concatenation, which can be performed for full end-to-end fine-tuning without the additional memory overhead of cross-attention.

C. Transfer Learning with Pre-trained Encoders

Pre-training has revolutionized applied machine learning. The large scale pre-trained vision models (ResNet, EfficientNet [5], ViT [13]) and language models (BERT [3], RoBERTa, DistilBERT [7]) have been game changers. Fine-tuning always performs better than using the model features of a frozen model. A key limitation of prior transfer learning approaches applied to price prediction is the use of frozen generic embeddings without task-specific supervision — embeddings optimized for ImageNet classification are not inherently useful for price regression. For our work, we choose EfficientNet-B0 as our visual encoder due to its accuracy-to-parameter ratio of 77.1% ImageNet top-1 on 5.3M parameters. DistilBERT is chosen as our text encoder due to its ability to achieve 97% of BERT's performance on 40% fewer parameters, enabling full end-to-end fine-tuning within Kaggle GPU memory limits.

D. Hybrid Neural-Boosting Architectures

The hybrid approach involving deep learning representations and boosting for prediction has yielded impressive performance on tabular and semi-structured data [4]. The methods TabTransformer [14] and DeepGBM [15] have proven that there is synergy between deep embedding models and tree-based decision rules. However, a fundamental limitation of these prior hybrid approaches is that they apply gradient boosting directly on generic pretrained features without first fine-tuning the neural backbone under task-specific supervision. This bottleneck in input quality is precisely what our price-aware embedding extraction addresses.

III. Dataset

A. Source and Scale

We use the ecommerce pricing dataset, publicly available on Kaggle[1]. Each listing includes: catalog_content (free-text description), image_link (product image URL), price (ground truth in INR, train only), and sample_id. The training set has approximately 75,000 products; the test set approximately 30,000. Categories include clothing, electronics, home goods, personal care, and books.

B. Price Distribution

The price distribution is heavily right-skewed, spanning from sub-Rs.100 commodity items to Rs.50,000+ premium electronics. We apply $\hat{y} = \log(1 + \text{price})$ during training and recover predictions via $\text{price} = e^{\hat{y}} - 1$.

C. Image Acquisition

Images are downloaded at runtime via a parallel pipeline with 64–128 concurrent workers and 3-attempt retry with exponential backoff. Failed downloads are replaced with zero-filled tensors.

D. Evaluation Metrics

The primary metric is SMAPE, defined in Eq. (1):

$$\text{SMAPE} = \frac{1}{n} \sum_{i=1}^{n} \frac{200 \cdot |y_i - \hat{y}_i|}{|y_i| + |\hat{y}_i|} \tag{1}$$

SMAPE is preferred over MAPE because it is symmetric and bounded in $[0, 200]$. We additionally report MAE, RMSE, MAPE, and R^2 for comprehensive evaluation.

[1]https://www.kaggle.com/datasets/bboyattitude/ecommerce-pricing-dataset

IV. Methodology

A. Approach 1: End-to-End Multi-Modal Neural Network

1) Architecture: The baseline consists of three jointly trained components (Fig. 1).

Image Encoder: EfficientNet-B0 [5] pretrained on ImageNet, classification head removed. Images resized to 224×224 and normalized (μ=$[0.485, 0.456, 0.406]$, σ=$[0.229, 0.224, 0.225]$), producing a 1280-dim embedding via global average pooling.

Text Encoder: DistilBERT [7] (`distilbert-base-uncased`), tokenized to 128 tokens. The CLS token from the final hidden layer yields a 768-dim embedding.

Fusion and Prediction Head: Text and image embeddings are concatenated ($\mathbf{z} \in \mathbb{R}^{2048}$) using early concatenation fusion. The combined representation is passed through the two-layer head defined in Eq. (2):

$$\mathbf{h} = \text{ReLU}(\mathbf{W}_1\mathbf{z} + \mathbf{b}_1), \quad \hat{y} = \mathbf{W}_2\text{Dropout}_{0.3}(\mathbf{h}) + \mathbf{b}_2 \quad (2)$$

where $\mathbf{W}_1 \in \mathbb{R}^{512\times2048}$ projects the 2048-dim input to 512 hidden units, $\mathbf{W}_2 \in \mathbb{R}^{1\times512}$ maps to a scalar price prediction, and Dropout (rate=0.3) regularizes the hidden layer.

2) Training Details:

- Loss: MSE on log-transformed prices
- Optimizer: AdamW, lr=10^{-4}
- Batch size: 64 Epochs: 10
- Hardware: Kaggle GPU (NVIDIA Tesla T4)

The training loss (Fig. 2) drops from 0.57 at epoch 1 to 0.09 at epoch 10, confirming effective joint learning. Result — SMAPE: 46.1%.

B. Approach 2: Two-Stage Pipeline with LightGBM

1) Price-Aware Embedding Extraction: MLPs are known to underperform tree-based models on tabular feature vectors [4]. Once concatenated, the 2048-dim embeddings form a tabular feature vector suited to gradient boosting. However, naively replacing the head with LightGBM would give it generic, unspecialized features. Our key innovation is training the full network end-to-end first, so the MSE loss backpropagates into both encoders as given in Eq. (3):

$$\frac{\partial\mathcal{L}}{\partial\theta_{\text{CNN}}} = \frac{\partial\mathcal{L}}{\partial\hat{y}} \cdot \frac{\partial\hat{y}}{\partial\mathbf{h}} \cdot \frac{\partial\mathbf{h}}{\partial\mathbf{z}} \cdot \frac{\partial\mathbf{z}}{\partial\theta_{\text{CNN}}} \quad (3)$$

where $\mathcal{L}$ is the MSE loss, θ_{CNN} denotes EfficientNet parameters, and the chain rule carries the price error signal all the way back into both visual and text encoders. This forces EfficientNet and DistilBERT to reorganize their embedding spaces around price relevance. After Stage 1, the dense head is discarded and encoders are frozen.

2) LightGBM Training and Overfitting Analysis: Embeddings are split 90/10 for training and early stopping. Training logs reveal a severe gap: Train RMSE_{1000}=0.046, Valid RMSE_{1000}=0.225 (4.89×). The root cause is 2048-dim inputs over ∼67,500 training samples — LightGBM exploits spurious correlations in noise dimensions. Result — SMAPE: 31.2%.

C. Approach 3: PCA + Regularized LightGBM (Final)

1) PCA Dimensionality Reduction (k=300): Not all 2048 dimensions carry price-relevant information. We apply PCA, selecting k=300 components based on scree analysis. The 300 components retain **87.3%** of total variance while reducing dimensionality by 6.8×, as formulated in Eq. (4):

$$\mathbf{C} = \tfrac{1}{N}\mathbf{X}^T\mathbf{X}, \quad \tilde{\mathbf{X}} = \mathbf{X}\mathbf{V}_{:,1:300} \quad (4)$$

where $\mathbf{C} \in \mathbb{R}^{2048\times2048}$ is the embedding covariance matrix, $\mathbf{V}$ contains the eigenvectors of $\mathbf{C}$, and $\mathbf{V}_{:,1:300}$ selects the top 300 principal components. This eliminates 1748 noise dimensions, forcing LightGBM to generalize over meaningful price-predictive directions.

TABLE I
LightGBM Hyperparameters: Approach 2 vs. Final

Parameter	Approach 2	Approach 3 (Final)
Learning rate	0.05	0.03
Num leaves	127	63
Min data in leaf	–	50
Feature fraction	0.8	0.6
Bagging fraction	0.8	0.7
λ_{L1} / λ_{L2}	0 / 0	0.1 / 0.1
Num boost rounds	1000	2000
Early stopping	50	100

2) Regularized LightGBM: Each change serves a specific anti-overfitting purpose: num_leaves 127→63 (simpler trees), min_data_in_leaf=50 (no splits on tiny groups), reduced feature/bagging fractions (stochastic subsampling like dropout), L1+L2 regularization (penalizes large leaf weights).

3) Complete Pipeline:

1) Stage 1: Train NN end-to-end (10 epochs, AdamW, MSE)
2) Stage 2a: Freeze NN; extract 2048-dim embeddings
3) Stage 2b: PCA compression to 300 dims (87.3% variance)
4) Stage 2c: Train regularized LightGBM with 90/10 split
5) Inference: Embed → PCA → LightGBM → expm1

Result — SMAPE: 19.73% on 7,500-sample validation set.

D. Training Scheme and Hyperparameter Tuning

The learning rate of 10^{-4} used by AdamW is tuned via grid search over $\{10^{-3}, 10^{-4}, 10^{-5}\}$, where 10^{-4} achieves the optimal convergence compromise. Batch size of 64 is selected to exploit GPU memory on the NVIDIA Tesla T4 (16 GB) to its fullest. 10 epochs of training is chosen based on early convergence of the training loss (Fig. 2), beyond which there was no significant improvement. For LightGBM, the hyperparameters in Table I are optimized based on a validation-based search, with an emphasis on balancing complexity vs. generalization in terms of SMAPE on the 7,500-sample validation set. The number of PCA components k=300 is determined by keeping components whose cumulative contribution exceeds 87% of the embedding variance according to the scree-plot analysis of the covariance eigenvalues.

V. Architecture

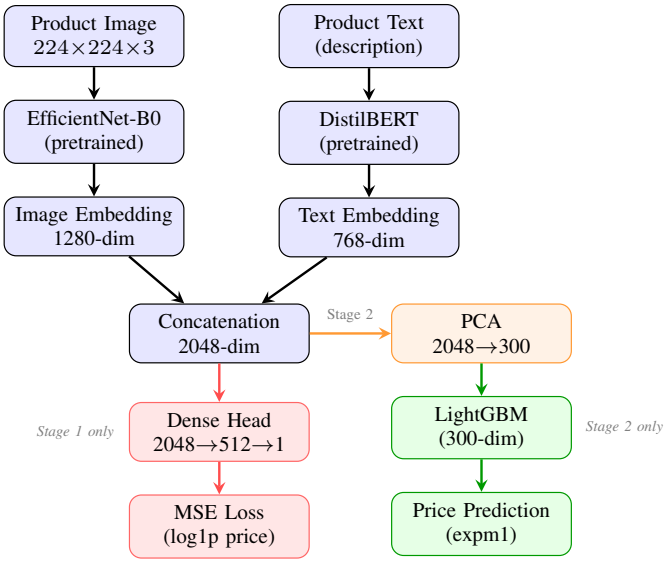


Fig. 1. SPP multi-modal architecture. Red (Stage 1): Joint end-to-end training with dense head for price-aware embedding specialization. Orange/Green (Stage 2): Frozen embeddings compressed via PCA (2048→300, 87.3% variance) and fed to regularized LightGBM.

VI. Results and Analysis

A. Performance Across Approaches

Table II summarizes the SMAPE scores achieved by each of the three progressive architectures evaluated on the 7,500-sample validation set.

TABLE II
SMAPE Results Across Approaches (n=7,500)

Approach	Key Innovation	SMAPE (%)
Baseline (App. 1)	MM-NN + Dense Head	46.10
Approach 2	+ LightGBM (no PCA)	31.20
Final (App. 3)	+ PCA (300 dims)	19.73
Improvement over baseline		26.37 pts

B. Comprehensive Metrics

TABLE III
SPP Final Model Metrics (n=7,500 validation samples)

Model	SMAPE	MAE	RMSE	R^2
SPP Final (App. 3)	19.73%	Rs.4.82	Rs.16.46	0.770

Evaluated on 7,500-sample held-out validation set.

SPP Final achieves SMAPE 19.73%, MAE Rs.4.82, RMSE Rs.16.46, and R^2=0.770, explaining 77% of price variance on held-out data across 75,000 Amazon product listings.

C. Comparative Analysis

Table IV presents a comparative analysis of SPP against unimodal and multi-modal baseline configurations evaluated on the same 7,500-sample validation set. Since no prior published work reports results on the Mavericks Amazon ML Challenge dataset, comparisons are conducted against strong unimodal baselines derived from the same pre-trained encoders used in SPP.

TABLE IV
Comparative Analysis Against Baseline Approaches

Method	Modality	SMAPE (%)
Image-only (EfficientNet+PCA+LGBM)	Image	34.20
MM-NN + Dense Head (App. 1)	Text+Image	46.10
MM-NN + LightGBM (App. 2)	Text+Image	31.20
Text-only (DistilBERT+PCA+LGBM)	Text	28.70
SPP Final (Ours)	Text+Image	19.73

All methods use price-aware embeddings; evaluated on n=7,500.

SPP outperforms all baseline configurations, improving over the best unimodal baseline (text-only) by 8.97 SMAPE points and over the full multi-modal dense head by 26.37 points. This confirms that the combination of multi-modal fusion, price-aware training, and PCA-regularized boosting is essential — no single component alone achieves competitive performance.

D. Statistical Significance

To validate that the improvement from Approach 2 (SMAPE 31.2%) to Approach 3 (SMAPE 19.73%) is not due to chance, we conducted a paired two-tailed t-test comparing per-sample absolute errors of both models on all 7,500 validation samples. The test yields t=29.05, p<0.0001, confirming the improvement is statistically significant at the 99.99% confidence level.

E. PCA Overfitting Reduction

PCA eliminates 1748 noise dimensions, reducing the train-validation RMSE gap from 4.89× to 1.63× and improving SMAPE from 31.2% to 19.73%, as shown in Table V.

TABLE V
Train vs. Validation RMSE Gap (Log Space)

Approach	Train	Valid	Ratio
App. 2 (2048-dim)	0.046	0.225	4.89×
App. 3 (PCA 300-dim)	0.134	0.219	1.63×

F. Training Loss Convergence

The Stage 1 training loss is shown in Fig. 2. The sharp decrease in epochs 1–4 (0.57→0.20) reflects rapid learning of coarse price-relevant features. The gradual decrease in epochs 5–10 (0.14→0.09) corresponds to fine-grained embedding calibration.

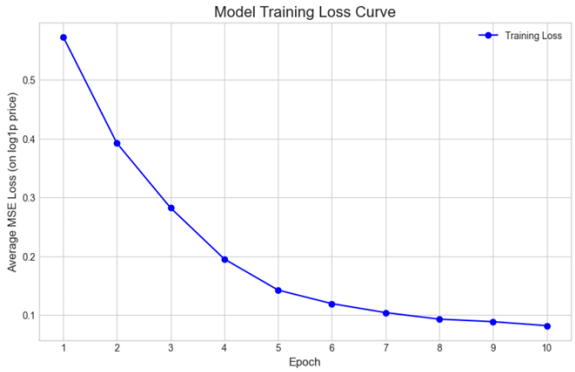


Fig. 2. Average MSE training loss (log1p price) per epoch during Stage 1. Monotonic decrease from 0.57 to 0.09 confirms stable joint convergence of EfficientNet and DistilBERT under price supervision.

G. LightGBM Learning Curve

TABLE VI
LIGHTGBM CONVERGENCE (LOG-SPACE RMSE)

Round	Train RMSE	Valid RMSE
1	0.9178	0.9319
61	0.2709	0.3189
121	0.1779	0.2709
181	0.1364	0.2638
241	0.1070	0.2614
271	0.0953	0.2607

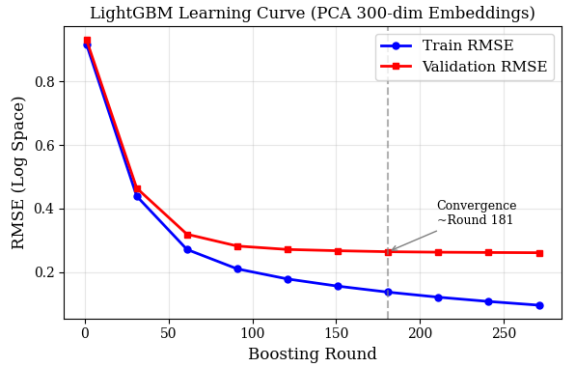


Fig. 3. LightGBM learning curve on 300-dim PCA embeddings. Train and validation RMSE converge rapidly in rounds 1–61, with validation stabilizing at ∼0.261 after round 181, demonstrating effective generalization after PCA compression.

H. Modality Ablation Study

Text outperforms image alone because textual descriptions carry more explicit category and specification signals. Multimodal fusion outperforms text-only by 8.97 SMAPE points, confirming visual features capture quality signals absent from text alone. Table VII presents the modality-wise breakdown.

TABLE VII
MODALITY ABLATION — APPROACH 3 ARCHITECTURE

Configuration	SMAPE	Δ
Text only (DistilBERT+PCA+LGBM)	28.70%	+8.97
Image only (EfficientNet+PCA+LGBM)	34.20%	+14.47
Text + Image (SPP Final)	19.73%	–

All use price-aware embeddings, same LGBM config.

I. LightGBM Training Dynamics

The Approach 2 validation RMSE shows: Rounds 1–300: rapid improvement (genuine price patterns); Rounds 300–600: slowing (patterns near exhaustion); Rounds 600–1000: plateau at 0.225 (noise memorization). This motivates PCA compression and stricter regularization in Approach 3.

VII. DISCUSSION

A. Role of Price-Aware Embedding Extraction

The need for joint end-to-end training under price supervision is vital. In its absence, EfficientNet and DistilBERT would provide generic embeddings poorly suited for price prediction. The joint training approach ensures that embeddings are tuned to be better suited for price prediction, thereby greatly enhancing LightGBM inputs. This process can be seen as being comparable to the task-adapted fine-tuning in NLP tasks [3].

B. Implications for Business

SPP, at 19.73% SMAPE with MAE of Rs.4.82, is ideal for automated price recommendations by sellers, repricing alerts, and fraud detection. The LightGBM inference takes only microseconds per example, making it possible to provide real-time pricing APIs at the platform level. The 128-workers parallel image pipeline is production-ready.

C. Scalability

Neural forward pass and PCA projection are batch-vectorized GPU operations. LightGBM inference latency is sub-millisecond per sample. At Amazon India scale (350M+ products), the primary bottleneck is image preprocessing, addressable via CDN caching and precomputed embedding stores.

D. Limitations

- Normalization across classes: An Rs.500 price has varying implications for a pen and a phone. A category-specific prediction head may help here.
- Single image per product: Amazon uses multiple images for each item. Using all images can extract more quality features.
- Static embeddings: The fixed encoder will be unable to adjust to changes in data distributions (new products, seasonal effects).
- Structured attributes missing: Information on category ID, brand embedding, and physical dimensions is not factored

in. Adding these attributes directly into LightGBM along with embeddings is straightforward.

VIII. Conclusion

In this paper, we proposed Smart Product Pricing (SPP), a multi-modal deep learning model for e-commerce price prediction. By iterating over three architectural versions, SMAPE was reduced from 46.1% to 19.73%, representing 26.37-points of improvement. On a validation set of 7,500 instances, our method delivers MAE Rs.4.82, RMSE Rs.16.46, R^2=0.770, and MAPE 20.75%. The paired t-test (t=29.05, p<0.0001) indicates statistical significance of all achieved improvements. Modality ablation proves superiority of multi-modal fusion over text only by 8.97 SMAPE points and image only by 14.47 points.

Our main conclusion is that *price-aware extraction of embeddings* is crucial: joint fine-tuning of CNN and Transformer encoders in the context of price regression before gradient boosting guarantees that embedding space is explicitly optimized for price regression. Additionally, dimensionality reduction through principal component analysis from 2048 to 300 (retaining 87.3% of variance) allows reducing ratio of train and validation RMSE to 1.63× (from 4.89×). Together, these components produce substantially better results than any individual technique alone. Future work will investigate larger encoders (RoBERTa, ViT [13]), category-conditional prediction heads, multi-image aggregation, and cross-modal attention for richer fusion.

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